

Detecting verifiability and bullshit in French Political Discourse with classical and deep learning methods

Jean-Lou Valeau

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Abstract

The goal of this project is to use NLP techniques to automatically detect political vagueness in French legislative election documents. We use Frankfurt's concept of bullshit, which consists in an indifference to accuracy, to classify political texts in two classes. Class 0 texts are verifiable, concrete propositions, while Class 1 texts are bullshit, vague, imprecise statements that do not commit the politician who wrote it to anything. We manually annotate approximately 1000 excerpts from the 1993 French legislative elections to train our models, and our dataset achieves a Fleiss kappa of 0.87 across the three annotators. We first train classical machine learning baselines, i.e. Logistic Regression, Linear SVM and Random Forests using TF-IDF representations and specific custom features that we built to discriminate bullshit texts from verifiable ones. These baseline models all reach a F1-macro of 0.77. However, these models have a poor Class 0 precision of 0.63-0.65, so they often misclassify bullshit texts as verifiable concrete texts. We then switch to using a CamemBERT model, which yields a F1-Macro of 0.80 and a Class 0 precision of 0.83. We analyze the errors of this model and show that the main causes of misclassification are excerpts combining verifiable and bullshit elements, as well as excerpts that are ambiguous and could be labeled both as 0 and 1. When applying our models to data from the 1973 and 1981 elections, the performance of all models decreases because of the shifts in distribution of political vocabulary and of the variation of class distribution of texts over the 1970s, 1980s and 1990s.

1 Introduction

"Political language [...] is designed to make lies sound truthful and murder respectable, and to give an appearance of solidity to pure wind", George Orwell declared in 1946. By definition, a politician must convince people to vote for him. To do so, he or she can use concrete arguments, or rely on rhetoric, charm, and smooth talk, which is often easier and less risky than presenting the harsh truth and a complex reasoning. In a world where populism rises and public trust decreases, it is therefore important to quantify the "bullshit" and empty promises made by politicians. Beyond the democratic stake, it is also an interesting and challenging topic in NLP.

In this project, we define verifiability as concrete propositions (class 0), by opposition to political vagueness (class 1), which we assimilate to the "bullshit" concept from the philosopher H. Frankfurt. Bullshit is a mode of discourse characterized by an indifference to precision. The bullshitter does not care about the accuracy of his assertions: his only goal is to produce a desired effect on his audience. In the context of political discourse, bullshit specifically aims at avoiding accountability while maintaining authority. By using abstract ideas, appropriate keywords and an operationally empty language, politicians can convince the voters without ever proposing any concrete proposition that can be measured. For instance, when discussing the economy, a politician can either be concrete and say that they *"will cut the corporate tax rate by 5% for the smallest companies starting next January"*. Or they can bullshit and state that they *"are committed to developing a pro-business ecosystem to revitalize our economic attractiveness"*.

Classifying bullshit and verifiability using NLP is more difficult than it may first look for 3 reasons. (i) First, it goes beyond simply looking for numbers, dates and quantitative information in the text. A statement like *"you've been ripped off for 5 years by the current government who raised taxes by 8%"* contains quantifiable information, but it is an attack on an opponent, not a personal commitment. On the contrary, a concrete commitment can be expressed without quantification: *"we will close all coal plants by the end of the mandate"*. (ii) Another difficulty is that political language is stylistically and grammatically homogeneous. So it is more

difficult to identify unambiguous features that discriminate between bullshit and concrete information, as explained in [1], who specifically focused on detecting fake news. (iii) Moreover, the task of annotating political content is very subjective and depends on the political view of the annotator, which creates noise in the labels and makes the task harder, as shown by [2, 3].

Hence, our goal in this project is to classify bullshit and verifiability in French legislative election campaign documents. Although the use of NLP has already been exploited extensively to analyze various political topics [4, 5], the automatic detection of bullshit and verifiability in political discourse remains underexplored, especially in French. To address this topic, our methodology is the following: we annotate manually 1000 political texts from the 1993 French legislative elections, according to a verifiability criterion: class 0 texts are defined as concrete and verifiable, class 1 texts are defined as bullshit and vague. We train classical machine learning baselines and compare their results to transformer models. We then keep the method with the highest F1-macro score and use it to predict bullshit and verifiability on out-of-sample data from the 1973 and 1981 data.

Our contributions are the following:

1. We present a new annotated corpus of French political speech from the 1993 legislative elections, of approximately 1000 datapoints. On the 199 texts annotated by all three annotators, we obtain a Fleiss's kappa of 0.87, which is robust and ensures there is not too much noise in the annotations.
2. We train classical benchmark models such as logistic regression, SVM and random forest using the TF-IDF approach, and we build a set of specific features that we designed to discriminate between bullshit and verifiability. We find that the baselines (TF-IDF plus logistic regression and SVM) reach a F1-macro score of 0.77.
3. We compare these classical benchmark to the Transformers deep-learning based approach and analyze the strengths and weaknesses of each approach. We find that the CamemBERT model beats the benchmarks and yields a F1-macro score of 0.8.
4. We apply our models to texts from other legislative elections and analyze the failures and limits of our approach.

The rest of this report is organized as follows. In section 2, we do a literature review and present relevant work and concepts for our project. In section 3, we present the dataset that we built, the annotation procedure and we do exploratory analysis on the data. In section 4, we train models and analyze their results. In section 5, we study the performance and limits of our procedure when applied to out-of-sample data from other elections. Finally, in section 6, we conclude our work and open with future directions of research.

2 Related work

Political text analysis with NLP techniques has been explored a lot to answer several questions. [4] introduced the foundations of automatic political text analysis by proposing a set of fundamental principles to ensure rigor when applying NLP to political science; in particular, their "validate, validate, validate" rule highlights the fact that researchers must constantly compare the algorithm's output against human interpretation to prove that the algorithm actually predicts what it is supposed to predict. This rule is especially relevant for our project and we'll use it in section 5. Building on these fundamental principles, NLP methods have been applied to a wide range of tasks in political science, such as ideology classification or sentiment analysis, or argument mining.

Argument mining is somehow related to our project. This field aims at identifying argument structure within texts. As explained in the survey by [6], its goal is to find various rhetorical techniques within a reasoning text, e.g. assumptions, arguments, examples or conclusion. The main difference with our work is that we do not attempt to decompose the structure of statements, but rather to discriminate between verifiable and bullshit statements. Another related field is automated fact-checking and more generally, fake news detections. [7] introduced ClaimRank, a model designed to detect claims in political discourse that should be fact-checked. The question of detecting fake-news was treated in [8], who introduced a new dataset of political lies. But once again, this approach is focused on identifying potentially false or dubious claims, not on predicting verifiability of the content. Some NLP work also focuses on detecting vague language. In particular, [9] proposed an

approach to detecting vagueness in legal and policy detection related to privacy policies. This work shows that vagueness as opposed to concreteness is a linguistic concept that can be quantified. However, their approach is specific to privacy policies and not applicable at all to political text.

Hence, the specific analysis of verifiability vs. bullshit and vagueness in political material is a relatively underexplored field, especially on French texts. The closest work related to our project that we could find is [10], who predicted pledge specificity in Australian election documents. They annotated thousands of political sentences on a 7-level scale, going from purely rhetorical pledges ("*we will put our country first*") to verifiable commitments ("*we will invest \$1 billion to support X*"). They implemented deep ordinal regression models to learn this multi-class classification models, but their dataset was very unbalanced, with two thirds of sentences belonging to the lowest class of the scale. As a result, the performance of their models was not very satisfying, because of the large number of classes. This motivates our simplified setting, in which we switched from a 7-classes classification problem to a binary classification problem. This makes the problem more tractable given our small dataset.

Regarding the algorithms: text classification in NLP was initially approached with bag-of-words representations such as TF-IDF combined to traditional machine learning algorithms such as logistic regression or SVM [11]. Nowadays, deep-learning based approaches are often more efficient, since the transformers architecture and the attention mechanism allows models to learn contextual dependencies between words [12]. Pretrained models for French language such as CamemBERT [13] are especially convenient for NLP classification tasks because their weights are off-the-shelf, the model only needs to learn a classification layer, and it provides excellent performances in many cases. However, in some cases, especially on high dimensional, low sample datasets, traditional TF-IDF based models can still outperform transformers which may overfit: this motivates introducing TF-IDF based models as baselines in our work.

3 Data, annotation procedure and exploratory analysis

3.1 Annotation procedure

Split of leaflets into individual excerpts. We built our dataset using leaflets from the 1993 French legislative elections. Each of these leaflets contains a series of statements that we have split into individual excerpts. The goal of this segmentation step was to isolate propositions with a self-contained meaning. As a result, The excerpts vary in length: some texts are a single sentence if the statement is clear enough on its own, but others texts are made of several sentences if the idea in the text depends on the previous sentences. For instance:

- a statement like "*We will increase the minimum wage by 5% and index it to inflation to protect the purchasing power of the most vulnerable workers*" can be treated as a single excerpt;
- the following statement contains a similar proposition, but it is formulated in several sentences: "*Rising prices have been eroding the purchasing power of the French people for several months now. It is therefore necessary to index wages to inflation. In particular, the poorest workers are hardest hit by inflation. I therefore propose to increase the minimum wage by 5%*". Hence, these 4 sentences are treated as a single excerpt as each individual sentence is directly related to the others.

Labeling of excerpts. In total, we annotated 961 excerpts. Each excerpt was assigned a binary label:

- Class 0 (verifiable and concrete texts) was assigned when the text contained at least one statement specific enough to be checked against facts based on the text alone. For instance, a statement like "*We will hire 10,000 new teachers over the next three years to resolve the crisis in the national education system*" contains a number, a timeframe, and a concrete action — all of which make it empirically verifiable after the end of the mandate of the politician.
- Class 1 (vague, non-verifiable bullshit texts) was assigned when a text does not commit to anything that could be tested or measured, i.e. when it could not be labeled with the Class 0. The 5 most frequent cases in which a text is labeled with the Class 1 is when:

- the statement is an abstract affirmation related to the vision statement of the candidate and to rhetorical padding, e.g. *"We must give France the means to face the challenges of tomorrow by remaining true to our values"*;
- the statement is too general or not accurate enough, e.g. *"We are going to resolve the crisis in the national education system by supporting teachers"*. We'll see in section 4.3 that these sentences sometimes yield false positive;
- the statement is a call to action or to mobilization of the reader, e.g. *"vote for our party on Sunday"* or *"each voice is important, don't let the others decide for you"*;
- the statement is related to presenting the candidate, e.g. *"I've been a town councillor for the past 10 years so I'm aware of the problems of our region"*.

Homogeneity of annotations. Many excerpts fall clearly into one category or the other, however some cases are harder to classify. These examples are the ones that cause more errors in the predictions of our models, as discussed in section 4. As shown by [3], annotating manually political texts can be very subjective and result in noisy, unreliable labels that vary with the annotator; this is all the truer as the three annotators in our group did not necessarily share the same political opinions. To ensure that annotations were not too biased, 199 texts were annotated by the three of us to form a shared dataset. On these 199 texts, the Fleiss Kappa was of 0.87, which ensures that there is not too much divergence between annotators. When looking at the texts on which we disagreed, the annotation of the text was always open to debate, and it was difficult to classify it clearly as 0 or 1.

In addition, we computed the proportion of Class 1 texts in the texts annotated by each annotator. We found out that the proportion of Class 1 texts was respectively of 0.74, 0.65 and 0.69, which further reassured us regarding the homogeneity of our annotations and the application of our manual classification rule.

Political orientation of the excerpts. To analyze the evolution of verifiability vs. bullshit by political party through time, each text was also annotated with its political orientation, depending on the party associated with the text. Concretely, each text was categorized under one of the following political orientations: Radical Left, denoted as GR (Gauche Radicale), Left, denoted as G (Gauche), Center, denoted as C (Centre), Right, denoted as D (Droite), Radical Right, denoted as DR (Droite Radicale), Unclassifiable, denoted as I (Inclassifiable).

For simplicity, we decided to group far-left and far-right parties into the categories of radical left and radical right. Although these are not exactly the same, there were too many political orientations, and we did not have enough data points for certain categories if we had separated extreme parties from radical parties. We also labeled parties that are not aligned with the classical political scale as unclassifiable.

3.2 Data exploration

The distribution of texts across the political spectrum is uneven, as shown by figure 1: 41% come from the left, 22.6% from the right, 18.3% from the radical left, 9.2% from the radical right, 6% from independent parties, and 2.9% from the center. The left, right, and radical left together account for over 82% of the corpus. Texts from the radical right, the center, and independent parties are underrepresented, which reflects the marginal role of these movements in France in the 1990s.

The proportion of verifiable texts is not uniform across political orientations. According to our annotations:

- The political orientation with the lowest proportion of concrete (Class 0) texts is the right wing (D): this proportion is of 15% while the average of Class 0 texts on the whole dataset is of 31%.
- The political orientation with the largest proportion of Class 0 texts is the group of unclassifiable parties (I), with a proportion of 67.4%. However, we have very few texts for this political orientation so this figure should be considered carefully. We can also comment that we have few observations for the far-right wing (DR) and the center wing (C), so their proportion of Class 1 texts may not be relevant.
- The left wing and far-left wing are respectively associated with a Class 0 texts proportion of 32.9% and 31.8%, which matches the whole dataset average of 31%.

These figures highlight a potential risk regarding classification: since the distribution of classes is different across each political orientation, the predictive power of our classifiers could come from correlations between linguistic

features and political orientation, rather than from verifiability itself. This risk is all the more significant as the frequent words in Class 0 texts of the left-wing and right-wing orientations are very different (see figures 4,5 in appendix). Frequent words in Class 0 for the left-wing are *"work, local, housing, national, local"* while frequent words in Class 1 for the right-wing are *"retirement, company, immigration, taxes"*.

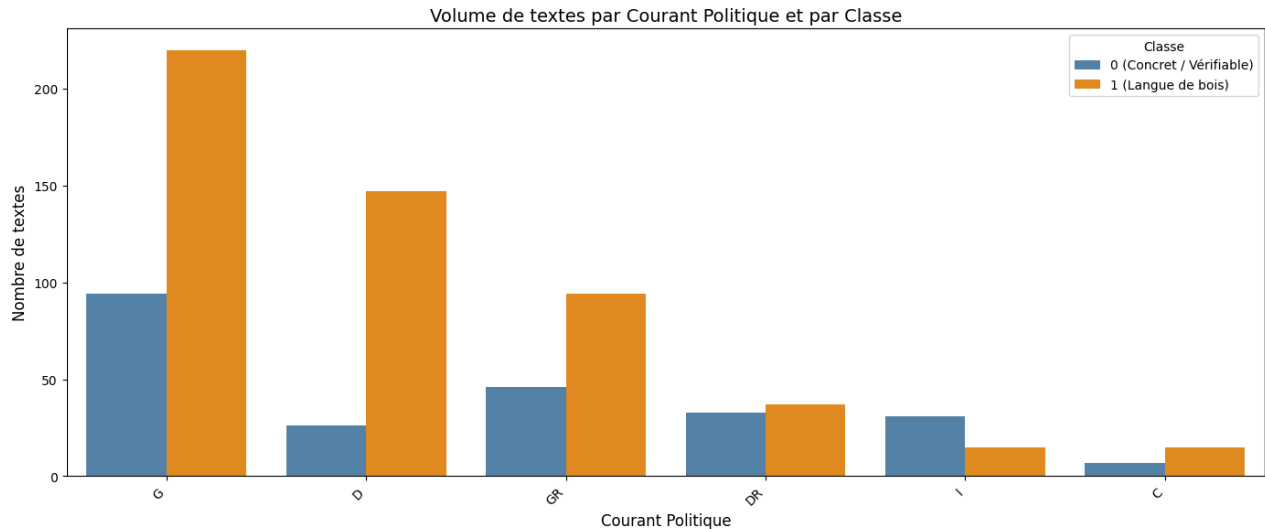


Figure 1: Number of texts per political orientation and per class

When producing wordclouds of frequent words in each class and across the whole dataset, we get a confirmation that our labeling procedure yields meaningful differences between the two classes.

The WordCloud of frequent words in Class 0 (see the blue wordcloud in figure 2) shows that Class 0 is dominated by words related to governance, administration, and quantification. This includes numerical and economic terms (*"salary, income, million, 000"*), spatial and administrative vocabulary (*"housing, local, rural, european"*), and action-oriented nouns and verbs (*"creation, reform, project, construction"*). These are the kinds of words one would expect in concrete, measurable policy proposals.

Class 1 (see orange wordcloud in figure 2) contains words related to the rhetoric of election, eg vocabulary that refer to the election itself rather than to a specific program. We find generic political terms that, during the labeling phase, were mostly observed in introductions or conclusions of leaflets (*"vote, party, candidate, democracy, Sunday 21st"* = the election day). We also find abstract concepts (*"solidarity, progress, trust"*), as well as modal verbs expressing intention rather than commitment (*"want, have to"*).

Interestingly, some of the most frequent words appear in both classes (see green wordcloud in figure 2): *"unemployment, employment, work, environment, country"*. These are the central themes of the campaign, and their presence in both classes suggests that what distinguishes verifiable from non-verifiable discourse is not the topic itself, but the way it is addressed, either through concrete proposals or through vague ideas.

4 Models and implementation

4.1 Implementation of baselines

4.1.1 Data pre-processing and features engineering

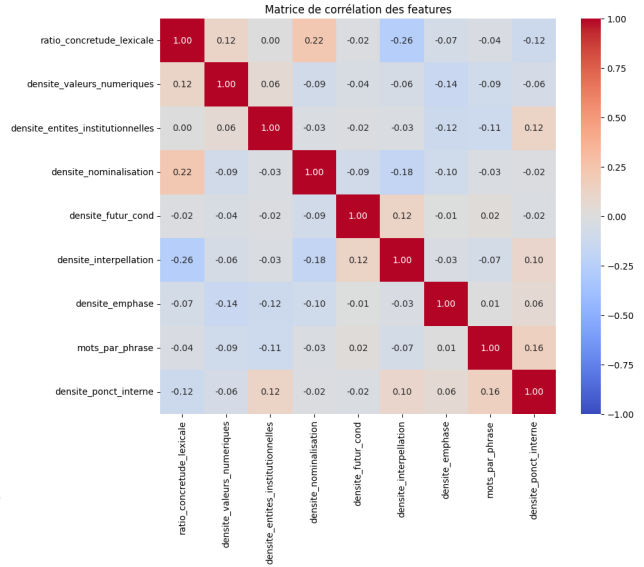
Before training any model, we apply several preprocessing steps and build a set of custom features designed to capture linguistic differences between the two classes.

Text preprocessing. Since our corpus was digitized through OCR, the texts contain broken words across line breaks, bullet points, weird spacing. We clean these using regular expressions, then process the texts with spaCy's French pipeline to lemmatize the text. For instance, *"taxes"* and *"taxation"* will be mapped to the same token *"tax"*. We remove words that appear in fewer than 3 documents: these are often proper nouns such



Feature selection. To verify that these features carry discriminative information, we apply a Mann-Whitney U test on each feature between Class 0 and Class 1. All features yield a p-value below 0.05, except sentence length and institutional entity ratio, which we discarded from the final feature set. We also checked the correlation matrix of the features that we built (see figure 4.1.1). All features were weakly correlated so we kept them all.

Scaling and class weighting. The numerical features are standardized using a RobustScaler, which relies on the median and interquartile range rather than the mean and standard deviation, making it less sensitive to outliers present in the data. Since the dataset is imbalanced, we apply class weighting during training, assigning a higher weight to Class 0 samples in the loss function to prevent the model from being biased toward the majority class.



4.1.2 Training of baselines

Logistic regression, Linear SVM and random forests are trained on the train set of pre-processed texts and features. The parameters are selected using a stratified 5-fold cross-validation. Since the dataset is high-dimensional, of size 765×1503 , ℓ_2 -penalization was added to logistic regression and SVM.

Table 1 summarizes the test set performance of our baseline models and compares it to the performance of the Transformers model. The majority class classifier, which always predicts Class 1, achieves an accuracy of 0.64 but an F1 Macro of only 0.39. This shows that accuracy should not be the key metric when classes are not balanced and it motivates the use of F1 Macro as the main evaluation metric.

All the trained models reach a similar F1 Macro of 0.77, which is a huge improvement over the majority baseline: this shows that the models are actually learning something and are not overfitting. Logistic Regression and Linear SVM produce very similar results. This makes sense as both are linear models with the same ℓ_2 regularisation, trained on the same features. Random Forest also has a F1 Macro of 0.77. Its difference with the two other models is that Random Forest achieves the highest recall (0.78) but the lowest precision (0.63), which implies that it detects more verifiable texts, but at the cost of a false positive rate.

The main limit of all our models is the precision of Class 0 (0.63–0.65). This means that when a model predicts a text as verifiable, it is wrong about one third of the time. As a result, one third of the texts that are labeled by the model as verifiable are actually bullshit texts. This is likely because these bullshit texts contain the same thematic vocabulary as texts of verifiable propositions. This is aligned with the observation in section 3 that political vagueness uses the vocabulary of concrete propositions (e.g. *emploi*, *environnement*, *pays*, *chômage*,..., see green wordcloud 2 of common words in the two classes). Since TF-IDF representations cannot represent this distinction, this performance ceiling motivates the use of contextual models such as CamemBERT, which are better suited to discriminate between identical words used in different contexts.

4.2 Implementation of a Transformers model

We fine-tune a pretrained CamemBERT model [13] on our binary classification task. The text preprocessing rules that we chose for the Transformers model are much simpler than those used for the benchmark models: we only clean whitespace and missing values, without applying lemmatization or stop word removal. This is motivated by the fact that Transformers rely on contextual representations that are designed to work with raw text directly, so that preprocessing the text too much could remove useful information that the model could exploit. We fine-tune the model for 3 epochs using a learning rate of 2×10^{-5} and a batch size of 4.

Table 1 compares CamemBERT with the baseline models. CamemBERT achieves an F1 Macro of 0.80, a 3-point improvement over all baselines. We also see a serious improvement over baselines regarding Class 0 precision, which is at 0.83 for the Transformers model, and which was capped at 0.65 for the baselines. This

Model	Class 0 (Verifiable, concrete)		Class 1 (Vague, bullshit)		Accuracy	F1 Macro
	Precision	Recall	Precision	Recall		
Majority Class Classifier	0.00	0.00	1.00	0.64	0.64	0.39
Logistic Regression (L2)	0.65	0.75	0.88	0.82	0.80	0.77
Linear SVM (L2)	0.65	0.75	0.88	0.82	0.80	0.77
Random Forest	0.63	0.78	0.89	0.79	0.79	0.77
CamemBERT	0.83	0.63	0.85	0.94	0.84	0.80

Table 1: Comparison of trained baseline models and CamemBERT on the test set. Best results per column are in bold.

implies that the Transformers model is better at discriminating concrete texts from bullshit texts with the same thematic vocabulary as concrete texts, e.g. *chômage*, *emploi*, *environnement*. This shows that contextual representations can capture differences in how words are used, not just which words are present. Yet, the overall improvement provided by these contextual representations remains limited as our dataset is small: the training dataset is made of 765 points, which is not enough and causes a bit of overfitting when training the Transformers model.

However, there is a decrease of Class 0 recall: it was of 0.75-0.78 for the baselines and it drops to 0.63 with the Transformers model. The Transformers model is more conservative in its predictions of Class 0: when it predicts a text as verifiable, it is very often right, but it misses a larger proportion of verifiable texts: it means that using the CamemBERT model will reduce false positives rate on Class 0 but increases false negatives. On Class 1, recall reaches 0.94, meaning the model correctly identifies almost all vague texts.

From now on, we'll focus on the Transformers model as it is overall the one with the best results.

4.3 Errors analysis

We now examine two types of errors of the CamemBERT model: false positives (vague texts predicted as verifiable) and false negatives (verifiable texts predicted as vague).

False positives. Consider the following excerpts, labeled as Class 1 but predicted as Class 0:

- *"Réduire les impôts en mettant fin aux gaspillages de l'argent public et en supprimant progressivement l'impôt sur le revenu"*. The model is probably misled for two reasons:
 1. the programmatic form of the statement, which reads like a concrete policy measure, with an apparent mechanism (ending public waste, gradually removing income tax). However, these elements are not really verifiable: what is as *"gaspillage d'argent public"* is not defined, and *"progressivement"* does not commit to any timeline or quantified target. The text looks like a concrete proposal but it is in fact very vague.
 2. the words *"impôts"*, *"revenu"*, used in the statement are frequently used in class 0 (see figure 2), and the programmatic structure of the sentence probably misled the model and prevented it from understanding the context in which these words are used.
- *"Adapter l'éducation afin de donner à chaque jeune une chance d'acquérir une formation débouchant sur un emploi"*. The model is again misled for the same reasons:
 1. the sentence also uses action verbs and has a programmatic structure, but the proposal remains abstract. There is no way to empirically verify whether *"each young person"* has been given *"a chance."*
 2. the words *"formation"*, *"emploi"* are again frequently used in class 0.

False-positive texts are therefore caused by a combination of words frequently used in Class 0 and a structure resembling that of a concrete proposition. This concrete proposition structure prevents the Transformer from having a proper contextual representation for the high-frequency words in Class 0, and as a result, the model

classifies these Class 1 texts as Class 0.

False negatives. Consider the following excerpts, labeled as Class 0 but predicted as Class 1:

- *"Pour donner aux vins de l'Aude le prestige qu'ils méritent, une loi intelligente doit permettre la publicité qualitative de nos vins dans tous les médias"*. This text contains a concrete and verifiable legislative proposal, i.e. allowing wine advertising in media. However, it is associated with rethorical padding and bullshit words that are abstract and that could be associated with Class 1 text: *"le prestige qu'ils méritent"*, *"une loi intelligente"*.
- *"Nous sommes contre l'extension sauvage de la plateforme de Roissy, pour le maintien des structures et emplois sur Orly et Roissy et un développement à taille humaine et supportable"*. Similarly, this text contains a verifiable statement, i.e. it is against firing employees from the Orly and Roissy airports and it wants to regulate the development of Roissy airport. Yet, it is associated with vague, abstract formulations typical of Class 1 texts, e.g. *"développement à taille humaine et supportable"*, which may explain why it was predicted as Class 1.

Annotation ambiguity and cause of errors. Some of the models errors may not necessarily be mistakes, but rather reflect the ambiguity of some texts. The previous excerpts discussing cutting taxes and the one about Roissy and Orly airports are good examples. These statements contain a specific proposition, which makes a Class 0 label reasonable, but it also lacks any quantified target or timeline, which also justifies a Class 1 annotation. In such ambiguous cases, the annotators could disagree, and the model's prediction, even though it is considered as wrong according to our annotations, still predicts the text in a legitimate way. This is aligned with [3], who argues that label variation in subjective NLP tasks does not necessarily reflect annotation error but genuine ambiguity in the data. In our binary classification setting, this is all the truer as we have to annotate as 0 or 1 some texts that are in between and could be classified as both. To reduce this ambiguity, it would be relevant to switch to multiclass classification and to add a third class to annotate ambiguous excerpts.

5 Out-of-sample analysis and discussion of results

The final step of the project is to analyze the performance of our models on data of the 1973 and 1981 legislative elections. To do this, we additionally labeled approximately 250 excerpts from election leaflets from those years. The performance of all models decrease when evaluated on out-of-sample data from the 1973 and 1981 elections, as shown by figure 3. In particular, the F1 Macro of Logistic Regression falls from approximately 0.78 on the 1993 data to 0.66 on 1973 and 0.63 on 1981. CamemBERT remains the model with the best performance, but its F1 macro score also decreases on predictions on 1973 and 1981 data.

Two factors can explain this decrease of performance on out-of-sample data. First, the political vocabulary distribution changes significantly between the 1970s, 1980s and 1990s. As shown in Figure 5, words that appear frequently in the 1973 leaflets, such as *"expansion, gaspillage, inflation, dépense"* — are not used anymore the 1993 documents. This is caused by the difference between the political and economic context of the 1990s and of the early 1970s, marked by the end of the post-war boom and rising inflation. Similarly, the 1981 texts contains frequent terms like *"allocation, efficacité, désarmement"* that do not appear in 1993, because the political landscape has changed: the Cold War has ended and the country has been governed by a socialist government for a decade. These vocabulary shifts mean that the TF-IDF features learned on the 1993 training data become less informative when applied to earlier periods. Similarly, the contextual representations of CamemBERT become less relevant when key terms of the 1973 and 1981 elections are not contained in the 1993 distribution of words.

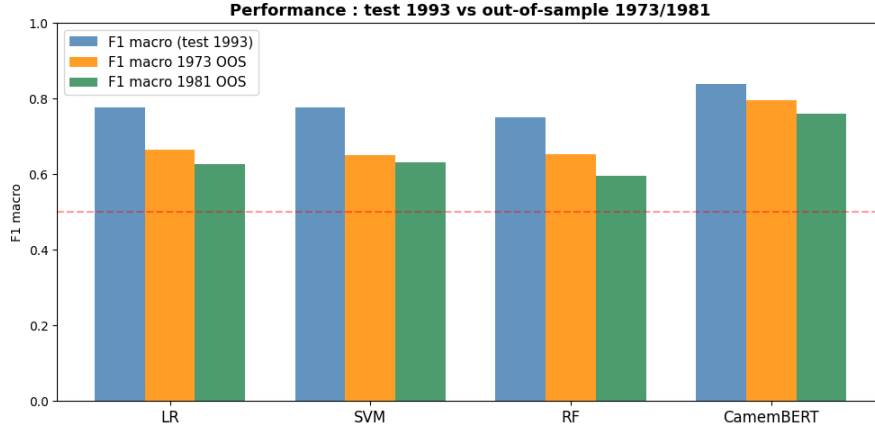
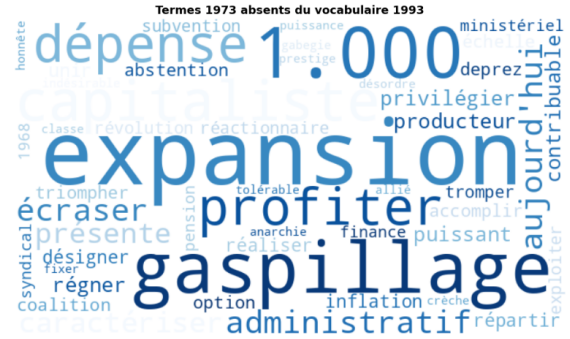


Figure 3: F1 macro score of the predictions of our models on 1973 and 1981 data

Second, the distribution of classes is not the same in 1973, 1981 and 1993. In the 1993 corpus, 69% of texts are labeled as Class 1 (bullshit), while this proportion is of 79% in 1973 and 73% in 1981, as shown by figure 7 in appendix. This shows that the rhetorical techniques used in politics change over time. Hence, it seems that candidates of the 1973 and 1981 campaign were using more heavily vague, bullshit language. This shift in class distribution creates a mismatch with the balance observed during training, which probably explains at least a part of decrease in the performance of the models.



6 Conclusion

In this project, we tried to automatically detect political bullshit and vagueness in French campaign documents. We annotated manually approximately 1000 excerpts from the 1993 French legislative elections, based on a verifiability criterion inspired from H. Frankfurt’s concept of bullshit. We showed that baselines such as Logistic Regression, Linear SVM and Random Forests with TF-IDF and custom features yielded a F1-Macro of 0.77. We managed to improve this score by fine-tuning a Transformers CamemBERT model with a F1-Macro of 0.80, mainly by increasing Class 0 precision from 0.65 to 0.83. This success illustrates the fact that contextual representations are efficient at discriminating vocabulary both used in vague bullshit statements and in verifiable propositions.

Our error analysis showed that most misclassification errors come from texts that incorporate both the characteristics of bullshit and verifiable excerpts. It also showed that some errors are caused by the ambiguity of the excerpts, which could be labeled both as Class 0 and Class 1, rather than by model failure. When applying our models to 1973 and 1981 French legislative elections texts, the performance of our models decreased because political vocabulary and distribution of classes changes across time periods. As a result, a limit of our approach is that a model trained on a single election can’t be directly used on texts of other elections.

This work could be extended in several ways. First, it would be interesting to apply the models to other types of elections, e.g. European elections or Presidential elections, to see in which case the model succeeds and fails. Moreover, our current model could be improved by training it on a larger dataset: our training dataset had only 765 observations, which is limited and increases the risk of overfitting, especially for deep learning methods. Finally, it would be relevant to introduce a third class in the labels, to classify texts that are ambiguous and that could be labeled both as Class 0 and Class 1.

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A Appendix



Figure 4: Most frequent words in class 0 texts of the right-wing



Figure 5: Most frequent words in class 0 texts of the left-wing

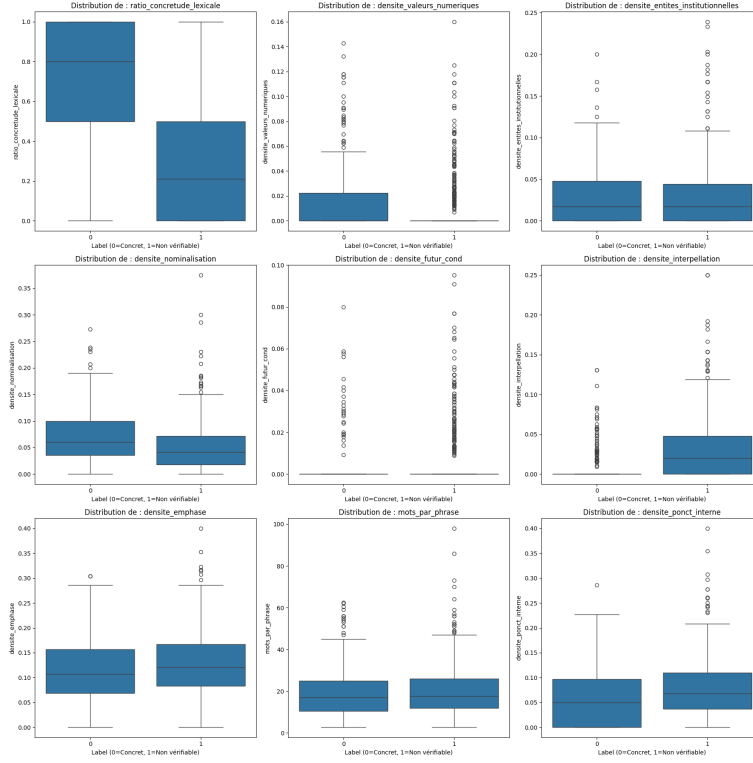


Figure 6: Boxplots of features distribution per class

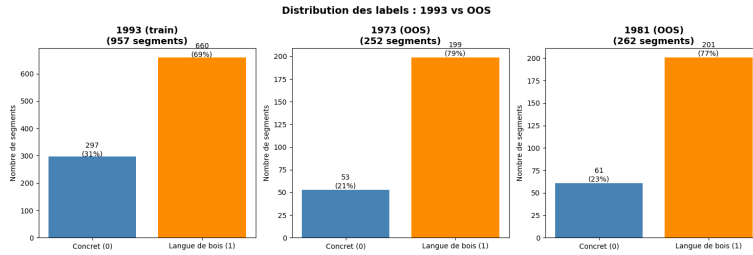


Figure 7: Distribution of classes on the texts of the 1973 and 1981 elections